Conditional Acceptance Calibration for Dynamic Medusa Decoding

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Abstract

Medusa speeds up language generation by drafting and jointly verifying future tokens. Its fixed- and dynamic-tree methods score candidate paths using products of separate prediction-head statistics, which may ignore dependencies. We will estimate token acceptance *conditional on the parent prefix being accepted* and use these estimates to select trees. We first test a count-based fixed-tree correction, then compare dynamic selection with MEDUSA+ and a feature-matched C2T-inspired classifier. The backbone and Medusa heads remain frozen.

1 Research Problem and Importance

Autoregressive models generate one token at a time. Speculative decoding [1] speeds up this process by verifying drafted tokens together; Medusa [2] uses additional heads to draft a tree of continuations. Larger trees may accept more tokens but require more verification work. Selecting useful branches under a fixed node budget is therefore essential. Original Medusa scores fixed-tree paths using products of head accuracies, while MEDUSA+ [7] builds dynamic trees using products of head probabilities. Both may misrank paths when token predictions depend on one another.

Research question: With the backbone and heads frozen, do parent-conditional acceptance estimates improve candidate selection over marginal-product scores and direct acceptance classifiers at the same tree budget?

2 Related Work and Research Gap

Hydra [3] makes draft heads depend on preceding tokens but changes the drafter. EAGLE-2 [4], Sequoia [5], and OPT-Tree [6] develop improved tree-selection methods. MEDUSA+ [7] applies dynamic trees to Medusa but retains marginal-product scoring. C2T [8] learns direct acceptance scores for tree pruning; TreeSpark [9] already studies calibrated parent-conditioned acceptance in another architecture. Thus, *conditional acceptance is not new*. Our contribution is a controlled study of whether post-hoc conditional scores help *frozen, non-autoregressive Medusa heads* without changing their predictions.

3 Proposed Method

1. Diagnose product-score error. We use greedy decoding only, accepting a candidate prefix if it matches the backbone's greedy continuation. For illustration, individual head accuracies of 0.80 and 0.75 produce a product estimate of 0.60; an observed joint success rate of 0.70 would reveal underestimation. We will measure the actual errors and their direction.

2. Calibrate conditional acceptance. Let S_v denote acceptance of the full prefix ending at node v, with parent u, and let x contain features available before verification. Estimate

$$q_v(x) = P(S_v \mid S_u, x), \qquad \widehat{P}(S_v \mid x) = \prod_{e \in \mathrm{path}(v)} \widehat{q}_e(x). \tag{1}$$

This is the probability chain rule, not an independence assumption; the estimated factors remain approximate. Since $\widehat{q}_e \in [0, 1]$, child scores cannot exceed parent scores. We will compare a **smoothed count table** indexed by head and preceding token ranks (one fixed tree) with **logistic regression** using ranks, confidence, depth, and parent–child interactions (dynamic trees). Conditional labels use only calibration examples whose parent prefix was accepted.

3. Select candidate trees. At each decoding step, expand high-scoring frontier nodes until reaching B nodes, always retaining ancestors. Using the *same candidates, selection procedure, and budget*, compare (i) MEDUSA+ marginal-product scoring, (ii) a C2T-inspired direct classifier with the *same features and capacity* as our logistic model, and (iii) our conditional path scores from Eq. (1).

4 Novelty and Significance

We do not claim that conditional acceptance or dynamic trees are new. We test whether calibration alone improves tree selection for frozen Medusa heads, and whether any gain survives online tree-building costs.

5 Evaluation and Feasibility

Compare the count-based fixed tree with original Medusa, and the conditional dynamic tree with MEDUSA+ and direct classification. Ordinary greedy decoding is the $1.0\times$ speed reference. The primary metric is **mean accepted tokens per verification step** at matched B; also report Brier score, tree-building overhead, end-to-end tokens/second, and agreement with greedy output. Calibrate on AlpacaEval prompts and evaluate on MT-Bench; HumanEval is optional. We plan to use the public **Vicuna-7B v1.3 Medusa-1** checkpoint with all weights frozen. We will check laptop GPU memory, try compatible 4-bit loading, or seek course compute. Trace-only results will not be reported as measured speedups.

6 Tentative Timeline

Week 1: GPU check and Medusa baseline. **Week 2:** Verification traces and error analysis. **Week 3:** Count, direct, and conditional estimators. **Weeks 4–5:** Matched-budget tree comparisons. **Weeks 6–7:** Timing, ablations, and report.

References

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